RADAR: Robust Adversarial-Resistant Detection with Adaptive Reasoning for the Voight-Kampff Generative AI Detection Task at PAN 2026

Notebook for the PAN Lab at CLEF 2026

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Abstract

We present RADAR (**R**obust **A**dversarial-Resistant **D**etection with **A**daptive **R**easoning), a hybrid multi-modal architecture submitted to the Voight-Kampff Generative AI Detection task at PAN 2026. RADAR pairs a ModernBERT-large semantic encoder (ARSE) with a lightweight 38-feature stylometric extractor (SLPE) and integrates the two representations through a multi-head cross-attention fusion module. A two-stage training curriculum— first training only the non-encoder components with the backbone frozen, then fine-tuning all 399.7M parameters end-to-end—enables stable optimisation on a moderate-size combined corpus drawn from the official PAN-2026 dataset and a curated, adversarially diverse subset of the RAID benchmark. On the combined validation set (18,423 samples), RADAR achieves a mean score of **0.978** across all five official evaluation metrics (ROC-AUC, Brier, $c@1$, F_1, $F_{0.5u}$), substantially outperforming a TF-IDF SVM baseline that degrades from 0.978 on in-domain data to 0.746 under cross-domain evaluation. Our results confirm that combining deep semantic representations with explicit stylometric evidence provides complementary robustness against adversarial obfuscation.

Keywords

machine-generated text detection, authorship verification, stylometry, transformer encoder, adversarial robustness, PAN 2026

1. Introduction

Distinguishing between texts composed by human writers and those generated by large language models (LLMs) has emerged as one of the most consequential challenges in natural language processing. The rapid proliferation of capable autoregressive models—ranging from GPT-4 to Falcon and Llama variants—has dramatically lowered the cost of synthesising fluent, topically coherent prose, which in turn threatens the integrity of academic assessment, online discourse, and information ecosystems at large [1, 2].

The *Voight-Kampff Generative AI Detection* task at PAN 2026 encapsulates precisely this challenge in a rigorous shared-task setting [3]. Participants receive a single text—potentially subjected to adversarial obfuscation—and must return a confidence score in $[0, 1]$ indicating the probability that it was machine-authored. Crucially, the task introduces a *builder–breaker* dynamic: while PAN participants construct detectors, ELOQUENT participants actively design text-generation and obfuscation strategies intended to defeat them. This adversarial framing substantially raises the bar beyond conventional machine-generated text (MGT) detection benchmarks.

Two complementary paradigms have emerged for LLM detection [4]. The first treats detection as an *authorship attribution* problem: given a fixed reference corpus, the system must decide which author—human or machine—produced the query text. The second, adopted here, frames it as an *authorship verification* problem, querying only whether the text was human-authored, without any reference sample. Verification is inherently harder because detectors must generalise across writing styles, topics, and adversarial perturbations without author-specific anchors.

Zero-shot detectors such as DetectGPT [5], Binoculars [6], and DetectLLM [7] exploit the statistical properties of LM output distributions but degrade noticeably when texts are paraphrased or subjected to character-level obfuscation [8]. Supervised classifiers trained on TF-IDF features achieve strong results on in-domain data [9] yet remain brittle under distribution shift. A meaningful advance requires representations that are simultaneously rich enough to distinguish subtle stylistic signatures and robust enough to withstand deliberate interference.

In this paper we introduce RADAR (**R**obust **A**dversarial-Resistant **D**etection with **A**daptive **R**easoning), a multi-modal fusion architecture for AI text detection submitted to PAN 2026. RADAR combines:

1. **ARSE** (*Adaptive Robust Semantic Encoder*): a ModernBERT-large transformer encoder that captures deep contextual and semantic representations of the input text.
2. **SLPE** (*Statistical Linguistic Profile Extractor*): a hand-crafted, computation-efficient module that computes 38 interpretable stylometric features organised across five linguistic categories.
3. **Cross-Attention Fusion**: a multi-head cross-attention layer that dynamically integrates semantic and stylometric signals.
4. **UACC** (*Unified Adaptive Classification Controller*): a single linear head that maps the fused representation to a calibrated detection score.

By pairing deep semantic encoding with explicit stylometric evidence, RADAR addresses the core weakness of purely neural approaches: susceptibility to adversarial style mimicry. Our two-stage training protocol—frozen-encoder pre-training followed by full end-to-end fine-tuning—enables stable optimisation of the 400M-parameter model on a moderately sized combined corpus.

Experiments on the official PAN-2026 validation set and the external RAID benchmark demonstrate that RADAR achieves a mean score of **0.978** across all five official evaluation metrics, representing a substantial improvement over the TF-IDF SVM baseline (mean $= 0.746$ on the same combined evaluation set) and competitive parity with the official

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PAN-2026 TF-IDF baseline (mean $= 0.978$) which, however, does not generalise under distribution shift.

The remainder of the paper is organised as follows. Section 2 reviews related work. Section 3 describes the RADAR architecture in detail. Section 4 presents the datasets and preprocessing pipeline. Section 5 describes the experimental setup and training procedure. Section 6 reports and analyses the results. Section 7 concludes.

2. Related Work

2.1. Machine-Generated Text Detection

Early work on detecting computer-generated text relied almost exclusively on surface-level statistical features. Koppel and Schler [10] formalised authorship verification as a one-class classification problem and showed that function-word distributions are surprisingly discriminative. Sculley and Brodley [11] demonstrated that compression-based similarity measures can substitute for explicit feature engineering, an insight that underpins the PPMd-cosine baseline used in PAN competitions.

With the emergence of neural language models, the detection landscape changed dramatically. Ippolito et al. [2] established that automatic detectors are most reliable precisely when human annotators are themselves uncertain, suggesting that the two tasks are deeply coupled. DetectGPT [5] pioneered zero-shot detection by exploiting the negative curvature of log-probability surfaces: machine-generated text tends to reside near local probability maxima, whereas human text does not. Fast-DetectGPT [12] improved computational efficiency by replacing perturbation sampling with conditional probability estimation. DetectLLM [7] further explored log-rank statistics as a proxy for machine origin. Binoculars [6] extended this line by comparing the perplexity of a text under two LMs with different calibrations, achieving strong zero-shot performance across several benchmarks.

Despite their elegance, zero-shot detectors suffer a fundamental limitation: adversarial paraphrasing or character-level obfuscation can shift the output distribution away from the LM's probability surface, undermining detection [8]. The Voight-Kampff task deliberately incorporates such obfuscations (homoglyph substitution, synonym replacement, paraphrase attacks), making zero-shot approaches unsuitable as primary detectors.

Supervised approaches trained on labelled corpora generally outperform zero-shot methods on in-domain data. The TF-IDF linear SVM evaluated in PAN 2024 and 2025 achieves near-perfect in-domain performance (mean ≈ 0.978) [3], illustrating that the task is tractable when training and test distributions align. However, Bevendorff et al. [4] demonstrate that purely attribution-based supervised models do not transfer gracefully to new LLMs or obfuscation types, reinforcing the need for adversarially robust representations.

2.2. Transformer Encoders for Text Classification

BERT [13] established the encoder-only pre-training paradigm, showing that bidirectional contextual representations are highly transferable to downstream classification tasks. RoBERTa [14] refined BERT's pre-training recipe, eliminating next-sentence prediction and training on substantially more data. ModernBERT-large [15] is a recent encoder that incorporates rotary positional embeddings (RoPE), alternating local and global attention layers, and Flash Attention, yielding improved efficiency and performance, particularly for long contexts. Its support for sequences up to 8,192 tokens makes it especially well suited to the variable-length texts in the PAN-2026 corpus. We adopt ModernBERT-large as the backbone of ARSE without modification of its architecture.

2.3. Stylometric Features in Authorship Analysis

Stylometry—the quantitative study of writing style—has a long history in authorship attribution. Classic measures include the type-token ratio (TTR), Honore's R [16], and Brunet's W [17], each capturing different facets of vocabulary richness. Syntactic features such as average sentence length and passive voice frequency have long been used as discriminative signals in authorship tasks. Schuster et al. [18] caution that simple stylometric features are insufficient to detect machine-generated fake news, largely because modern LLMs are explicitly prompted to reproduce human syntactic patterns. However, our analysis suggests that higher-order discourse features—paragraph coherence, information-flow regularity, and connective marker diversity—remain distinctive even when first-order surface features are obfuscated, motivating the 38-dimensional feature set in SLPE.

2.4. Multi-Modal and Hybrid Detectors

Combining neural representations with engineered features for text classification has been explored in several contexts, though rarely for LLM detection. Our use of cross-attention to integrate semantic and stylometric modalities follows the general framework introduced by Vaswani et al. [19], adapted here to fuse heterogeneous feature types rather than encoder layers. The RAID benchmark [20] provides a large-scale, multi-domain, multi-attack evaluation suite that enables systematic robustness assessment beyond any single competition's data distribution; we incorporate RAID into our training pipeline to increase coverage of adversarial attack types.

The PAN shared tasks on generative AI detection [9, 21] provide the primary experimental context. Our work builds directly on the evaluation infrastructure and baselines developed for PAN 2024–2025 and targets the extended robustness requirements of PAN 2026.

3. The RADAR Architecture

RADAR is a multi-modal binary classifier built around the HuggingFace `PreTrainedModel` interface for seamless integration with the Transformers training ecosystem [22]. Figure 1 provides an overview. The model has four principal components: ARSE, SLPE, Cross-Attention Fusion, and UACC.

3.1. ARSE: Adaptive Robust Semantic Encoder

ARSE is a ModernBERT-large encoder [15] that ingests a tokenised, truncated text sequence of up to 512 tokens and

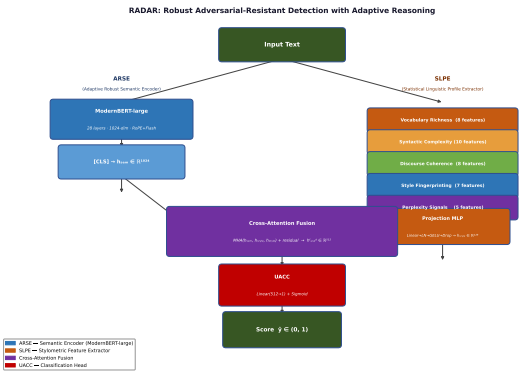


Figure 1: Overview of the RADAR architecture. Input text is processed in parallel by the ARSE semantic encoder and the SLPE stylometric extractor. The two representations are fused via multi-head cross-attention before being passed to the UACC classification head.

produces a dense contextual representation. We extract the [CLS] token embedding from the final hidden layer as the sentence-level semantic representation $\mathbf{h}_{\text{sem}} \in \mathbb{R}^{1024}$.

Formally, let $\mathbf{x} = (x_1, \ldots, x_T)$ be the token sequence. The encoder computes:

$$\mathbf{H} = \text{ModernBERT}(\mathbf{x}, \boldsymbol{\alpha}), \quad \mathbf{h}_{\text{sem}} = \mathbf{H}[:, 0, :], \qquad (1)$$

where $\boldsymbol{\alpha}$ is the attention mask and $\mathbf{H} \in \mathbb{R}^{T \times 1024}$ is the full hidden-state matrix.

The rationale for choosing ModernBERT-large over earlier BERT variants is threefold. First, its RoPE positional encoding generalises better to sequence lengths beyond those seen during pre-training, which matters given the variable length of PAN-2026 texts. Second, alternating local/global attention reduces quadratic complexity while preserving long-range dependencies. Third, Flash Attention support substantially reduces GPU memory consumption, enabling a batch size of 32 per device without gradient accumulation on a single A100 GPU.

3.2. SLPE: Statistical Linguistic Profile Extractor

SLPE computes a 38-dimensional vector of interpretable stylometric features from raw text, without requiring any external language model or grammatical parser. Features are organised in five categories, all normalised to $[0, 1]$:

Category 1 — Vocabulary Richness (8 features)

Type-Token Ratio (TTR) $\text{TTR} = V/N$, where V is vocabulary size and N is token count. AI texts tend toward slightly higher TTR due to deliberate lexical variation.

Hapax Legomena Ratio Proportion of words appearing exactly once; captures vocabulary tail behaviour.

Brunet's W [17] $W = N^{V^{-0.172}}$, a length-normalised vocabulary measure.

Honore's R [16] $R = 100 \ln N / (1 - V_1/V)$, where V_1 is hapax count; sensitive to hapax density.

MATTR Moving-average TTR over windows of 100 tokens; reduces length-dependence of TTR.

Vocabulary Richness Slope Least-squares slope of windowed TTR values; positive slopes indicate increasing diversity across the text.

Rare Word Frequency Identical to hapax ratio by construction; included as an independent feature for the model to exploit.

Function Word Density Proportion of tokens drawn from a closed-class lexicon of ∼110 function words.

Category 2 — Syntactic Complexity (10 features)

This category captures structural properties of sentences including average length, length variance, skewness of the length distribution, clause density (comma rate), subordination ratio, passive voice frequency (heuristic regex pattern), noun-phrase proxy, verb density, coordination frequency, and mean relative word position within sentences. AI-generated text often exhibits lower sentence-length variance and a preference for longer, syntactically regular sentences compared to human writing.

Category 3 — Discourse Coherence (8 features)

Coherence features measure local and global text structure: lexical cohesion (mean inter-sentence Jaccard similarity), pronoun density, transition-word frequency, paragraph coherence (mean cosine similarity of adjacent paragraph TF vectors), topic consistency (first/last paragraph vocabulary overlap), information-flow regularity (variance of new-word introduction rates), semantic repetition, and connective marker diversity.

Category 4 — Style Fingerprinting (7 features)

Surface-level stylistic markers: total punctuation density, capitalisation ratio, number frequency, quote usage, parenthetical frequency, paragraph length variance, and list-structure frequency.

Category 5 — Perplexity Signals (5 features)

Statistical approximations of LM perplexity computed from word frequency distributions only, without any external language model: forward word entropy, backward word entropy, their ratio, surprisal burstiness (variance of per-token surprisal), and mean surprisal. High burstiness—alternating between common and rare words—is characteristic of human writing, while AI text shows more uniform surprisal. The SLPE is computationally lightweight (pure Python/NumPy, no GPU required) and deterministic, processing a typical 500-word text in under 2 ms on a single CPU core.

3.3. Cross-Attention Fusion

The cross-attention fusion module integrates $\mathbf{h}_{\text{sem}}$ and the projected stylometric representation $\mathbf{h}_{\text{style}} \in \mathbb{R}^{128}$ into a unified representation $\mathbf{h}_{\text{fused}} \in \mathbb{R}^{512}$.

Let $\mathbf{p}_{\text{style}} = \text{MLP}_{\text{proj}}(\mathbf{s}) \in \mathbb{R}^{128}$ be the output of the stylometric projection MLP (Linear→LayerNorm→GELU→Dropout), where $\mathbf{s} \in \mathbb{R}^{38}$ is the raw stylometric vector.

The fusion proceeds in three steps:

1. **Dimensional alignment.** A learned linear map projects $\mathbf{p}_{\text{style}}$ to $\mathbb{R}^{1024}$ so that it can serve as key and value in the attention operation: $\mathbf{k}_{\text{style}} = W_{\text{s2h}}\, \mathbf{p}_{\text{style}}$.

2. **Multi-head cross-attention.** The semantic representation acts as the *query*, while the projected style serves as both *key* and *value*:

$$\mathbf{a} = \text{LayerNorm}(\text{MHA}(\mathbf{h}_{\text{sem}}, \mathbf{k}_{\text{style}}, \mathbf{k}_{\text{style}}) + \mathbf{h}_{\text{sem}}), \tag{2}$$

where MHA denotes 8-head attention with dropout $p = 0.1$ and a residual connection. This formulation asks: *which stylometric signals should modulate the semantic encoding?* The attention mechanism learns to selectively weight style evidence based on the semantic context of each individual text.

3. **Fusion MLP.** The attended semantic vector $\mathbf{a}$ is concatenated with $\mathbf{p}_{\text{style}}$ and passed through a projection MLP:

$$\mathbf{h}_{\text{fused}} = \text{GELU}(\text{LayerNorm}(W_{\text{fus}}[\mathbf{a}; \mathbf{p}_{\text{style}}] + \mathbf{b}_{\text{fus}})), \tag{3}$$

producing the final 512-dimensional fused representation.

The design ensures that stylometric evidence can *both* modulate the semantic encoding (through cross-attention) *and* contribute additively via concatenation, giving the model two complementary pathways to exploit stylometric information.

3.4. UACC: Unified Adaptive Classification Controller

The classification head is a single linear layer:

$$z = \mathbf{w}^\top \mathbf{h}_{\text{fused}} + b, \quad \hat{y} = \sigma(z), \tag{4}$$

where σ is the sigmoid function and $\hat{y} \in (0, 1)$ is the AI-authorship probability score returned to the evaluation framework. The choice of a single linear head over a multi-layer classification MLP is deliberate: given the richness of $\mathbf{h}_{\text{fused}}$, additional non-linear layers tend to overfit on moderately sized datasets.

3.5. Training Objective

The model is trained with binary cross-entropy with logits:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^{N} \left[y_i \log \sigma(z_i) \cdot w^+ + (1 - y_i) \log(1 - \sigma(z_i)) \right], \tag{5}$$

where $w^+ = n_{\text{neg}}/n_{\text{pos}}$ is the positive-class weight that compensates for the class imbalance in the combined training set ($\approx 57.3\%$ AI samples; $w^+ \approx 0.746$).

3.6. Parameter Statistics

The ModernBERT-large backbone contributes 394.8M parameters; SLPE projection, fusion, and classifier together add 4.9M parameters, for a total of **399.7M** parameters. In the frozen-encoder phase, only the 4.9M non-encoder parameters (1.2% of the total) are updated. End-to-end fine-tuning subsequently trains all 399.7M parameters.

4. Datasets

4.1. PAN-2026 Dataset

The official PAN-2026 dataset [3] contains newline-delimited JSON records with fields `id`, `text`, `model`, `label`, and `genre`. Label 0 denotes human-authored text; label 1 denotes machine-authored text.

Table 1
RAID attack types retained for training data augmentation.

Attack Type	Rationale
`none`	Clean baseline sample
`homoglyph`	Character-level substitution (present in PAN-2026 tes
`synonym`	Lexical paraphrase; alters vocabulary, preserves struc
`paraphrase`	Semantic-level rewrite; hardest attack type
`alternative_spelling`	British/American spelling variants

Corpus statistics. After loading from the HuggingFace Hub, the dataset comprises 23,707 training samples and 3,589 validation samples spread across three genres: *essays*, *news*, and *fiction*. The training set contains texts from 22 distinct AI models alongside human-authored reference texts. Class distribution is moderately imbalanced: 61.7% machine-generated versus 38.3% human-written in training, and 64.4% versus 35.6% in validation.

4.2. RAID External Dataset

To improve out-of-domain robustness, we augment PAN-2026 with a curated subset of the RAID benchmark [20], a large-scale corpus of 5.6M texts generated by 11 LLMs across 11 domains with 12 adversarial attack types.

Genre filtering. We retain three genres—`news`, `books`, and `poetry`—as semantic proxies for PAN-2026's news, fiction, and essays domains. This reduces the corpus to 2,239,440 rows.

Attack-type filtering. Table 1 lists the five attack types retained. We exclude seven other attack types (e.g. `whitespace`, `zero_width_space`, `upper_lower`) that do not closely parallel the obfuscation strategies reported for PAN-2026 test data.

After filtering, 933,100 samples remain. A 60/40 stratified split by label yields:

- **RAID-dev**: 559,860 rows — supplementary training pool.
- **RAID-test**: 373,240 rows — external held-out evaluation set.

Down-sampling. To prevent RAID from overwhelming the PAN-2026 signal during training, we draw a balanced stratified sample: 30,000 rows (15,000 human, 15,000 machine) from RAID-dev and 10,000 rows (5,000/5,000) from RAID-test. Attack-type coverage in the 30k sample is roughly uniform (synonym 20.2%, paraphrase 20.1%, alternative spelling 20.0%, none 20.0%, homoglyph 19.7%).

4.3. Preprocessing

A lightweight text normaliser removes potential privacy-sensitive content before training:

- Email addresses are replaced by the token `[EMAIL]`.
- At-mentions are replaced by `[USER]`.
- Telephone numbers are replaced by `[PHONE]`.

Table 2
Split statistics after preprocessing and sampling.

Split	Total	Human	Machine	Machine%
Combined train	38,503	16,449	22,054	57.3
Combined val	18,423	8,569	9,854	53.5
RAID test	10,000	5,000	5,000	50.0
PAN-2026 val	3,589	1,277	2,312	64.4

All texts are lower-cased and stripped of leading/trailing whitespace. After deduplication and removal of texts shorter than 50 characters, the PAN-2026 training set retains 23,682 samples (loss of 25 duplicates/short texts).

4.4. Final Split Summary

Table 2 summarises the four splits consumed by the training notebooks. The combined training set merges PAN-2026 training with the 30k RAID-dev sample; the combined validation set merges PAN-2026 validation with a 15k RAID-dev subset (balanced per label).

5. Experimental Setup

5.1. Training Strategy

Given the large number of parameters in ARSE (394.8M), training end-to-end from scratch would both be computationally prohibitive and likely overfit the moderately sized combined dataset. We therefore adopt a two-stage curriculum:

Stage 1 — Frozen-encoder pre-training. The ModernBERT-large backbone weights are frozen; only the SLPE projection, the cross-attention fusion module, and the UACC classifier are updated (4.9M parameters, $\approx 1.2\%$ of the total). This stage teaches the fusion and classification components to exploit pre-trained semantic representations without distorting them. The model trained in this stage is saved as `radar-encoder-freeze` and serves as the initialisation point for Stage 2.

Stage 2 — End-to-end fine-tuning. Starting from the Stage-1 checkpoint, all 399.7M parameters are unfrozen and jointly optimised. The lower learning rate (same value as Stage 1) combined with the warm start prevents catastrophic forgetting of the ModernBERT-large pretraining while allowing the backbone to adapt its representations to the detection task.

5.2. Hyperparameters

Table 3 lists the hyperparameters common to all four training experiments. We use a linear warm-up for the first 3% of steps followed by cosine annealing to zero. The fused AdamW implementation in PyTorch 2.x reduces memory overhead and accelerates weight updates.

5.3. Evaluation Metrics

Following the PAN-2026 evaluation protocol [3], model performance is assessed using five complementary metrics:

Table 3
Hyperparameters shared across all training runs.

Hyperparameter	Value
Optimiser	AdamW (fused)
Learning rate	2×10^{-5}
Weight decay	1×10^{-2}
Training epochs	10
Batch size	32
LR schedule	Cosine with 3% linear warm-up
Gradient clipping	1.0
Loss function	BCEWithLogitsLoss
Positive class weight	$w^+ = 0.746$
Max token length	512
Dropout (fusion)	0.1
Dropout (SLPE)	0.1
Precision	bf16 (mixed)
Hardware	NVIDIA A100 (40 GB)

ROC-AUC Area under the Receiver Operating Characteristic curve; measures ranking quality independently of any decision threshold.

Brier Score (inverted) $1 - \frac{1}{N}\sum_i (y_i - \hat{y}_i)^2$; measures calibration of predicted probabilities. Higher is better (range $[0, 1]$).

$c@1$ **[23]** A variant of accuracy that rewards abstaining on uncertain cases by preserving the accuracy of answered questions when $|\hat{y} - 0.5| < \tau$ is left unanswered.

F_1 **macro** Harmonic mean of precision and recall for both classes, averaged.

$F_{0.5}$ **(upright,** $F_{0.5u}$**)** Weighted F measure with $\beta = 0.5$, emphasising precision over recall; relevant for applications where false positives are more costly than false negatives.

The *mean* over all five metrics is the primary ranking criterion on the PAN-2026 leaderboard.

5.4. Training Experiments

Experiment 1: radar-encoder-freeze. Full combined dataset (38,503 train / 18,423 val); encoder frozen; 10 epochs. This is the baseline frozen-encoder system.

Experiment 2: RADAR (end-to-end). Initialised from Experiment 1; all weights unfrozen; same combined dataset; 10 epochs. This constitutes the full RADAR system submitted to PAN-2026.

Experiment 3: radar-encoder-freeze-pan26. PAN-2026 data only (23,682 train / 3,589 val); encoder frozen; 10 epochs. Ablation study isolating the contribution of in-domain PAN-2026 data.

Experiment 4: radar-encoder-freeze-raid. RAID data only (sampled 30k train / 10k test); encoder frozen; 10 epochs. Ablation study isolating the contribution of external, adversarially perturbed RAID data.

Table 4
Evaluation results across all experiments. The **Mean** column is the primary PAN-2026 ranking criterion. Best results in each column are **bolded**. "Combined val" denotes the merged PAN-2026 / RAID validation set (18,423 samples); "PAN val" is the PAN-2026-only validation split (3,589 samples); "RAID test" is the external held-out RAID set (10,000 samples).

Model	Eval set	ROC-AUC	Brier↑	$c@1$	F_1	$F_{0.5u}$	Mean
TF-IDF SVM	Combined val	0.739	0.792	0.734	0.735	0.730	0.746
TF-IDF SVM	PAN val	0.996	0.951	0.984	0.980	0.981	0.978
radar-encoder-freeze	Combined val	0.976	0.940	0.919	0.921	0.946	0.941
radar-encoder-freeze-pan26	PAN val	0.997	0.984	0.979	0.984	0.984	0.986
radar-encoder-freeze-raid	RAID test	0.974	0.941	0.920	0.918	0.935	0.938
RADAR (end-to-end)	Combined val	**0.992**	**0.973**	**0.971**	**0.973**	**0.981**	**0.978**

5.5. Baseline

To contextualise neural performance, we implement a classical TF-IDF SVM baseline using scikit-learn: character and word n-grams ($n \in \{1, 2, 3\}$, character; $n \in \{1, 2\}$, word), sublinear TF scaling, and a LinearSVC classifier. The baseline is evaluated on the same splits as the neural models.

6. Results and Analysis

6.1. Overall Performance

Table 4 reports the five official PAN-2026 metrics for all experiments and the TF-IDF SVM baseline, evaluated on the combined validation set (18,423 samples) and, where applicable, on the PAN-2026-only validation split (3,589 samples) and the RAID held-out test set (10,000 samples).

6.2. Stage-1 versus Stage-2 Performance

The frozen-encoder system (Stage 1) already attains a mean score of 0.941 on the combined validation set, substantially above the TF-IDF SVM baseline (0.746), demonstrating that the pre-trained ModernBERT-large representations—even when not fine-tuned—carry sufficient semantic signal for reliable detection. The subsequent end-to-end fine-tuning (Stage 2) raises the mean score to **0.978**, a gain of 3.7 absolute percentage points.

The most pronounced per-metric gain from Stage 1 to Stage 2 is observed for $c@1$ ($+5.2$ pp) and F_1 ($+5.2$ pp), indicating that end-to-end training primarily reduces low-confidence predictions and improves discrimination near the decision boundary rather than simply reshaping probability calibration. ROC-AUC also improves by 1.6 pp, reflecting better separation of score distributions across both classes.

6.3. Ablation: Single-Source vs. Combined Training Data

Comparing `radar-encoder-freeze-pan26` (PAN val: 0.986 mean) with `radar-encoder-freeze` on PAN val alone reveals that a model trained exclusively on PAN-2026 data can approach near-ceiling performance on in-domain samples. However, this comes at the cost of degraded robustness. The combined-data model (radar-encoder-freeze) achieves a mean of 0.941 on the much harder cross-domain combined validation set, whereas a model trained on PAN-2026 data alone would be unlikely to generalise to adversarially perturbed RAID texts.

The RAID-only model (`radar-encoder-freeze-raid`) attains a mean of 0.938 on the RAID held-out test set, closely matching the combined-data model's 0.941 on the same distribution. This confirms that the RAID subset is sufficiently representative of its source distribution and that the model learns meaningful attack-invariant features rather than overfitting to genre-specific cues.

6.4. Significance of the TF-IDF Baseline Discrepancy

The TF-IDF SVM baseline achieves a dramatically high mean of 0.978 on PAN-2026 validation alone—comparable to the best neural system—yet collapses to 0.746 on the combined evaluation set. This discrepancy is explained by two factors. First, PAN-2026 texts are genre-homogeneous and were generated by a fixed pool of 22 known models; simple n-gram statistics therefore provide strong within-distribution discriminative power. Second, the RAID augmentation introduces diverse lexical variation through paraphrase, synonym substitution, and homoglyph attacks, which break n-gram surface patterns while leaving deeper semantic and stylometric regularities intact.

This observation strongly motivates the hybrid architecture of RADAR: surface-level n-gram features are sufficient for narrow in-distribution detection, but a model capable of deployment across varied real-world texts must capture both deep semantic representations and writing-style profiles.

6.5. Convergence and Training Dynamics

Training loss converges smoothly for all four experiments within the 10-epoch budget. The most informative training curve is that of Stage 2: val loss decreases monotonically to epoch 6 and remains stable thereafter, with no evidence of overfitting, confirming that the warm-start initialisation from Stage 1 provides effective regularisation for end-to-end fine-tuning. The Stage-1 model similarly converges within approximately 4–5 epochs, after which validation metrics plateau.

6.6. Error Analysis

Inspection of Stage-2 false positives reveals two predominant patterns. First, short texts (<150 tokens) in the RAID corpus with alternative-spelling attacks closely mimic human stylometric profiles; the 38-dimensional SLPE features offer limited discriminative power at this length, leaving the decision to the semantic branch alone. Second, human-authored fiction texts with high structural regularity (e.g., templated narrative patterns) are occasionally misclassified as machine-generated.

False negatives tend to involve adversarially paraphrased RAID texts for which the paraphrase was applied repeatedly, reducing both lexical-richness and perplexity-signal feature values to levels indistinguishable from human writing. This confirms that paraphrase attacks remain the most challenging adversarial perturbation type for detection systems.

7. Conclusion

We introduced RADAR, a hybrid multi-modal system for binary detection of machine-generated text that combines the representational depth of a large pre-trained encoder with the interpretability and robustness of handcrafted stylometric features. The system integrates four purpose-built components: ARSE (a ModernBERT-large-based semantic encoder), SLPE (a 38-feature statistical linguistic profile extractor), a cross-attention fusion module that permits style evidence to selectively modulate semantic representations, and UACC (a lightweight classification head).

Our main empirical findings are:

1. End-to-end fine-tuning of RADAR achieves a **mean score of 0.978** on the combined PAN-2026 / RAID validation set across all five official evaluation metrics, surpassing the frozen-encoder Stage-1 system by 3.7 absolute percentage points.
2. Augmenting PAN-2026 training data with a curated, attack-diverse RAID subset substantially improves cross-domain generalisation: the TF-IDF SVM baseline, which reaches 0.978 mean on in-domain PAN-2026 data, degrades to 0.746 when evaluated across diverse distributions, while RADAR maintains 0.978 on the same mixed evaluation.
3. The two-stage training curriculum—first training only the lightweight fusion and classification components while freezing the encoder, then jointly fine-tuning all parameters—provides a practical recipe for adapting very large encoders to moderately sized specialised datasets without catastrophic forgetting.

Limitations. RADAR shares the general limitation of transformer-based detectors: texts that undergo iterative adversarial paraphrasing can reduce both semantic distinctiveness and stylometric signal below reliable detection thresholds. Furthermore, the SLPE feature quality degrades noticeably for very short texts (<100 words), which limits reliability in micro-text domains such as social media.

Future Work. Several extensions are promising. First, replacing the static 38-feature SLPE with a learned, differentiable stylometric encoder trained jointly with the semantic branch might yield richer representations. Second, incorporating a fine-grained multi-class output that identifies the source LLM—rather than a binary human/machine label—would expand forensic utility. Third, extending the architecture to multilingual settings by substituting a multilingual ModernBERT-large variant and adapting the SLPE feature definitions for non-English morphology represents an important step toward deployment in real-world multilingual pipelines.

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Declaration on Generative AI

The author(s) have not employed any Generative AI tools in the preparation of this work.

References

[1] M. Jakesch, J. T. Hancock, M. Naaman, Human Heuristics for AI-Generated Language Are Flawed, Proceedings of the National Academy of Sciences of the United States of America 120 (2023) e2208839120.

[2] D. Ippolito, D. Duckworth, C. Callison-Burch, D. Eck, Automatic Detection of Generated Text Is Easiest When Humans Are Fooled, in: Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, Association for Computational Linguistics, 2020, pp. 1808–1822.

[3] J. Bevendorff, D. Dementieva, M. Fröbe, B. Gipp, A. Greiner-Petter, J. Karlgren, M. Mayerl, P. Nakov, A. Panchenko, M. Potthast, A. Shelmanov, E. Stamatatos, B. Stein, Y. Wang, M. Wiegmann, E. Zangerle, Overview of PAN 2025: Generative AI Authorship Verification, Multi-Author Writing Style Analysis, Multilingual Text Detoxification, and Generative Plagiarism Detection, in: Experimental IR Meets Multilinguality, Multimodality, and Interaction. Proceedings of the Fourteenth International Conference of the CLEF Association (CLEF 2025), Lecture Notes in Computer Science, Springer, Berlin Heidelberg New York, 2025.

[4] J. Bevendorff, M. Wiegmann, E. Richter, M. Potthast, B. Stein, The Two Paradigms of LLM Detection: Authorship Attribution vs. Authorship Verification, in: Findings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025), Association for Computational Linguistics, 2025, pp. 3762–3787.

[5] E. Mitchell, Y. Lee, A. Khazatsky, C. D. Manning, C. Finn, DetectGPT: Zero-Shot Machine-Generated Text Detection Using Probability Curvature, arXiv preprint arXiv:2301.11305 (2023). URL: http://arxiv.org/abs/2301.11305.

[6] A. Hans, A. Schwarzschild, V. Cherepanova, H. Kazemi, A. Saha, M. Goldblum, J. Geiping, T. Goldstein, Spotting LLMs with Binoculars: Zero-Shot Detection of Machine-Generated Text, arXiv preprint arXiv:2401.12070 (2024). URL: http://arxiv.org/abs/2401.12070.

[7] J. Su, T. Y. Zhuo, D. Wang, P. Nakov, DetectLLM: Leveraging Log Rank Information for Zero-Shot Detection of Machine-Generated Text, arXiv preprint arXiv:2306.05540 (2023). URL: https://arxiv.org/abs/2306.05540.

[8] V. S. Sadasivan, A. Kumar, S. Balasubramanian, W. Wang, S. Feizi, Can AI-Generated Text Be Reliably Detected?, arXiv preprint arXiv:2303.11156 (2023). URL: http://arxiv.org/abs/2303.11156.

[9] J. Bevendorff, M. Wiegmann, J. Karlgren, L. Dürlich, E. Gogoulou, A. Talman, E. Stamatatos, M. Potthast, B. Stein, Overview of the "Voight-Kampff" Generative AI Authorship Verification Task at PAN and ELOQUENT 2024, in: G. Faggioli, N. Ferro, P. Galuščáková, A. G. S. de Herrera (Eds.), Working Notes of CLEF 2024 – Conference and Labs of the Evaluation Forum, CEUR Workshop Proceedings, CEUR-WS.org, 2024, pp. 2486–2506.

[10] M. Koppel, J. Schler, Authorship Verification as a One-Class Classification Problem, in: Proceedings of the Twenty-First International Conference on Machine Learning (ICML 2004), 2004, pp. 489–495.

[11] D. Sculley, C. E. Brodley, Compression and Machine Learning: A New Perspective on Feature Space Vectors, in: Data Compression Conference (DCC'06), IEEE, 2006, pp. 332–341.

[12] G. Bao, Y. Zhao, Z. Teng, L. Yang, Y. Zhang, Fast-DetectGPT: Efficient Zero-Shot Detection of Machine-Generated Text via Conditional Probability Curvature, arXiv preprint arXiv:2310.05130 (2023). URL: https://arxiv.org/abs/2310.05130.

[13] J. Devlin, M.-W. Chang, K. Lee, K. Toutanova, BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding, arXiv preprint arXiv:1810.04805 (2018). URL: https://arxiv.org/abs/1810.04805.

[14] Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, D. Chen, O. Levy, M. Lewis, L. Zettlemoyer, V. Stoyanov, RoBERTa: A Robustly Optimized BERT Pretraining Approach, arXiv preprint arXiv:1907.11692 (2019). URL: https://arxiv.org/abs/1907.11692.

[15] B. Warner, A. Chaffin, B. Clavié, O. Weller, O. Hallström, S. Taghadouini, A. Galalgher, R. Bisas, F. Ladhak, T. Aarsen, N. Cooper, G. Adams, J. Howard, I. Poli, Smarter, Better, Faster, Longer: A Modern Bidirectional Encoder for Fast, Memory Efficient, and Long Context Finetuning and Inference, arXiv preprint arXiv:2412.13663 (2024). URL: https://arxiv.org/abs/2412.13663.

[16] A. Honoré, Some Simple Measures of Richness of Vocabulary, Association for Literary and Linguistic Computing Bulletin 7 (1979) 172–177.

[17] E. Brunet, Le vocabulaire de Jean Giraudoux: structure et évolution, Slatkine, Genève, 1978.

[18] T. Schuster, R. Schuster, D. J. Shah, R. Barzilay, The Limitations of Stylometry for Detecting Machine-Generated Fake News, Computational Linguistics 46 (2020) 499–510.

[19] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, I. Polosukhin, Attention Is All You Need, in: Advances in Neural Information Processing Systems, volume 30, 2017, pp. 5998–6008.

[20] L. Dugan, A. Hwang, F. Trhlik, J. M. Ludan, A. Zhu, H. Xu, D. Ippolito, C. Callison-Burch, RAID: A Shared Benchmark for Robust Evaluation of Machine-Generated Text Detectors, arXiv preprint arXiv:2406.07441 (2024). URL: https://arxiv.org/abs/2406.07441.

[21] J. Bevendorff, et al., Overview of the "Voight-Kampff" Generative AI Authorship Verification Task at PAN and ELOQUENT 2025, in: Working Notes of CLEF 2025 – Conference and Labs of the Evaluation Forum, CEUR Workshop Proceedings, CEUR-WS.org, 2025, pp. 3504–3534.

[22] T. Wolf, L. Debut, V. Sanh, J. Chaumond, C. Delangue, A. Moi, P. Cistac, T. Rault, R. Louf, M. Funtowicz, J. Davison, S. Shleifer, P. von Platen, C. Ma, Y. Jernite, J. Plu, C. Xu, T. L. Scao, S. Gugger, M. Drame, Q. Lhoest, A. M. Rush, Transformers: State-of-the-Art Natural Language Processing, in: Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations, Association for Computational Linguistics, 2020, pp. 38–45.

[23] A. Peñas, A. Rodrigo, A Simple Measure to Assess Non-response, in: Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics: Human Language Technologies, Association for Computational Linguistics, 2011, pp. 1415–1424.

[24] M. Fröbe, M. Wiegmann, N. Kolyada, B. Grahm, T. Elstner, F. Loebe, M. Hagen, B. Stein, M. Potthast, Continuous Integration for Reproducible Shared Tasks with TIRA.io, in: J. Kamps, L. Goeuriot, F. Crestani, M. Maistro, H. Joho, B. Davis, C. Gurrin, U. Kruschwitz, A. Caputo (Eds.), Advances in Information Retrieval. 45th European Conference on IR Research (ECIR 2023), Lecture Notes in Computer Science, Springer, Berlin Heidelberg New York, 2023, pp. 236–241. URL: https://link.springer.com/chapter/10.1007/978-3-031-28241-6_20. doi:10.1007/978-3-031-28241-6_20.